

Car plate character extraction under complicated environment*

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Abstract - License plate recognition (LPR) system has many applications. Character extraction is a key step in LPR system. The proposed character extraction method in this paper is mainly according to plate's geometrical characteristic and supplementary with the scan of projection image. Subsequently, locate the plate character sequence, then do more jobs on false characters elimination and verify the validity of extracted characters. The above methods lead to an ideal character extraction result which leads higher recognition accuracy of the LPR system.

Keywords: LPR Character Extraction

1 Introduction

License plate recognition is necessary and effective way of vehicles monitor and management, License plate recognition system is widely used in some application systems. Such as toll road charge, city traffic management, intelligence traffic, etc.

There are three main processes in license plate recognition named: plate allocation, character extraction, character recognition. After plate allocation, plate characters are contained in the gray file. Firstly we transfer word the gray file to binary one. Then character extraction is to get each character out from the plate. The traditional algorithms are mainly based on vertical projection. The proposed method in this paper is according to the single connected areas in the binary plate file and then with the values of character width and character central position taken from connected areas, we can get char extraction result with the scan of projection image. Projection method is also considered to help to raise the character extraction accuracy.

Current LPR systems can be divided into two categories: Sensor trigger system and Portable system. When vehicle passes by, sensor on the system trigger the video capture system. Portable system use software trigger the video capture system, special software designed for vehicle capture is installed in portable system, it's the software that decided when to send current image to recognize. Both of the two kinds of systems have their

advantages as well as its shortcomings. With the help of affiliated hardware, the former always can get higher image quality, but it is comparative expensive. The portable system can be easily installed in any place. Now, Portable system is more widely accepted. However, the facultative video capture angle leads to the inclination of the image. Free capture distance leads to the large variation size of the plate area in the image captured. In all, the images captured by portable system are more difficult to recognize, the algorithm proposed in the article aims at these problems and does effective jobs to solve it, some novel ideas are introduced and ideal character extraction gotten.

2 Algorithm

2.1 Skew correction

The proposed input allocation image to the character extraction process is a binary plate file. See Figure. 1.



Figure 1. Plate binary file with inclination

Firstly we detect if there are inclination in the plate. Traditional inclination correction uses Hough algorithm, according to the edge of the plate so that calculate the incline angle of the plate. But in actually situation, the plate has not clear plate edge to detect sometimes, or the plate edge was not included in the allocation plate image. Therefore, we take the following method: labeling the connected areas, and record the area data in the following data structure:

```
struct Connected_Area{
    int width,height;
    int Numbersof0;
    int Numbersof1;
    struct ConnectedArea* next;
}
```

According to the connected area information, we can get the top and bottom position of each connected area, these position values can imitate two lines by the least-squares method, then calculate the angle of the two lines

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between horizontal direction, then inclination correction can be done.

After the inclination correction, we can extract characters from the plate area according these characteristics: Pixels number, Rate between width and height, Rate of foreground pixels and background pixels. More than two words need to be found to define the too lines

Algorithm I :

- (1) Labeling the connected areas;
- (2) Select ideal characters which meet the proposed characteristics just mentioned;
- (3) fgsgdfsgdfsgdf
- (4) Find the top point of each area; See figure2(a)(marked with 'x'), use least-squares method to imitate two lines named L0 and L1. The angle of L0 and L1 between horizontal direction are α and β , get its average value $\theta = (\alpha + \beta) / 2$. If $\theta > 30^\circ$, the plate are considered to be a false one. If $\theta < 3^\circ$, the inclination can be ignored as it has neglect affection to plate recognition;
- (5) Rotate the image file angle θ to its reverse direction, the result of the binary plate see Figure 2(c).

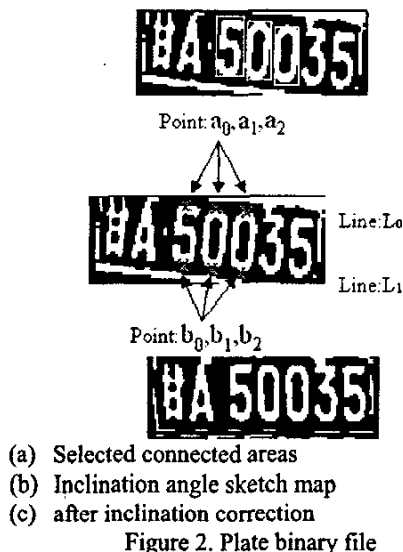


Figure 2. Plate binary file

2.2 Noise deletion

This step is to delete noise lies in vertical direction in the plate binary file, these noise deletion will lead higher system accuracy.

Select threshold $T1=15$, use scanning line method, begin from the center of the binary file and extends rightward, if there is 0 to 1 or 1 to 0 change between two neighboring pixels, the counter add 1, if the counter $\geq T1$, add the scan line by 1 pixel and do continue scan. If the total counter $< T1$ when one line scanning ended, the plate top edge is found. With the same method, we can find the plate bottom edge. See Figure 3.



Figure3 Plate binary file after noise deletion

2.3 Character extraction

Do vertical projection to the plate binary file, see Figure4(b). For better vision result, black line was dropped at the bottom of the binary file.

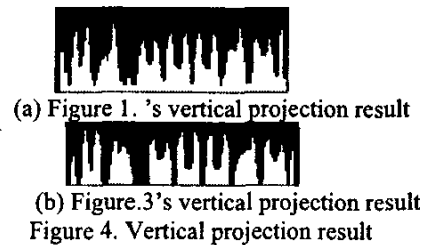


Figure 4(a) is the projection result of Figure1. As the noise exists, there are connections between characters in the projection file; Figure4 (b) is the projection result of Figure3. Obviously, the connection between characters are reduced in Figure4 (b), therefore, Figure4 (a) is abandoned, and Figure4(b) is chosen for further steps of character extraction.

According the Chinese plate style characteristics, characters in the plate have same width, which can be gained from the width of pre-extracted connected area, E.g. '5', '0'. At the same time, the pre-extracted connected area also give the character coordinate position in the plate. Now we can do character extraction with the combination with projection file and connected area given values. Note, the character center position here means the horizontal position, which is the character left edge add the half with of the character. The distance between two neighboring characters are called character center position distance.

The following steps are to do character extractions:

- (1) The pre-extracted connected area can give the vertical range of the plate characters. See h0 in Figure9.
- (2) The pre-extracted connected area can give the width of plate character (w_1) and character center position (d_0/d_1), see Figure9. According to plate geometry characteristics, all the characters have the same width, the last five characters have same character center position distance, d_0 is between 2nd character and 3rd character, d_1 is between other characters; $d_0 > d_1$, experiments show that $d_0 = \Delta t \times d_1, \Delta t \in [1.2, 1.3]$.
- (3) Scan the connected areas in h0 range in the plate file, integrating the projection file position information (Figure5), do area combination or area split.
- (4) Give the extraction result. The extraction result includes the characters position, and characters can be extracted accurately. See Figure6.



Figure 5. Scan the projection file in horizontal direction



Figure 6. Character extraction result (rectangle package shows it)

2.4 Character sequence location

Ideal plate allocation result give the accurate seven characters in the plate(see Figure1), traditional character sequence location method is based on the ideal plate allocation result, it's named standard pattern matching method. It takes the first character in the plate as the Chinese word, and the following six characters as alphabets and numbers. But in common situation, especially in portable LPR systems, plate area changed in large variation and always come with bad quality image, noise are often included in the allocated plate file(see Figure7). If we take the traditional character sequence location method, errors can't be avoided.

Now the character sequence location here is to find the accurate character begin position in the plate. You can see it's a key step in the process of character extraction.



(a)



(b)

Figure 7 Plate binary file



(a)



(b)

Figure8 Character extraction result before sequence allocation

Figure8 is the result after connected areas labeling. Noise area are included in the plate file, therefore, we need to do character sequence location. Analysis on the geometry of the Chinese vehicle plate, there are following characteristics:

- (1) Seven characters included;
- (2) Each character has the same width and height;
- (3) A round point between 2nd and 3rd character; First two characters distance and the last five characters have the same interval, d_1 ;
- (4) The interval between 2nd and 3rd character are comparable bigger, d_0 ;

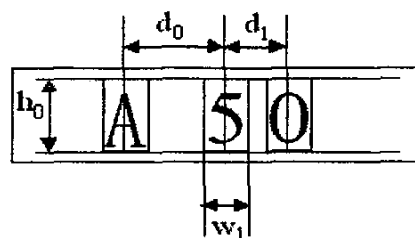


Figure 9. Geometry structure of the Chinese plate

According to the five characteristics mentioned, the most effective way is to find the point between 2nd and 3rd character. The unique point can give us the right sequence of the characters. But unfortunately, as there are serious pollution on Chinese vehicle plate, the small point is sometime missed. Therefore we have to find out other solutions. With the same reason, according to (5), d_0 is also unique position in the plate, if we can find the position of d_0 , the character sequence can be well allocated.

According to (2),(4),(5), following algorithm are got:
The algorithm process is as following:

- (1) Calculate each character center position, it is the character left edge add the half with of the character;
- (2) Calculate the interval between characters, stored in data array DistanceBetweenChar[];
- (3) Find the valid position biggest value in DistanceBetweenChar[];
- (4) Verify the validation of the position: if the following four intervals' $Er < T_2$, the position is valid, algorithm ends, otherwise, invalid this position, repeat step (3);
- (5) If no valid position, the plate is considered as a false one.

Er is calculated by following formula: Order the data in DistanceBetweenChar from smallest to biggest, get the middle one as Average. If d_0, d_1, d_2, d_3 are the following four intervals, then

$$Er = \sum_{i=0}^3 (average - d_i)^2 \quad (1)$$

Experiments show that, When $T_2=20$, ideal identification result can be gained. The noise in Figure8 can be eliminated. The accurate character sequence allocation result is in Figure10.



Figure 10 Plate result after character sequence allocation

2.5 Chinese word and rivets process

There is a Chinese word in the beginning of the Chinese vehicle plate, which is the abbreviation of each province/direct-managed-city. Some of them have separated structure, which introduces one word with several

connected areas. Of course, we must combine them or take other measures to avoid it. As we can get the accurate position of the character after the first Chinese word and the character interval d0, we can get character with and horizontal position. The vertical position of the Chinese character is the same as other characters.

In addition, there are always rivets in the plate, which lies in top and bottom of 2nd and 6th character. The two connected areas always taller than others, with the standard character vertical position and character height, rivet noise can be removed. Experiments show that this step is necessary and effective which leads to high system accuracy.

2.6 Plate character validity identification

In portable LPR systems, free video capture angle leads to that the plate of vehicles can be sheltered from other vehicles or bicycles. This kind of images will also be sent to the LPR system to recognize. It's clear that it's impossible to get the correct recognition result. So we must filter these images out. As we have mentioned, plate character characteristics including: pixels number, Rate between width and height, Rate of foreground pixels and background pixels. If the character doesn't meet these characters well or there are less than seven valid characters in the plate, the plate is identified to a false one.

3 Result

Figure11 shows the character extraction result of our method. These characters are required to be transferred to standard template. The template we use is 32*16 pixels. These characters can later be recognized by OCR system.

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Figure11 Character standardization result (white-black reversed)

Table 1 shows the accurate ration of our system.

Table 1. Experimental Results

Plate location	The Number of Images	Accurate Ratio	Time Cost(ims)
Xinjiang,Gansu Province	2501	96.94%	29
	12580	95.78%	36
	8401	98.62%	17
Zhejiang Province, Shanghai City	1242	95.90%	23
	18240	96.66%	27
	5080	98.38%	21
Total/Average	48044	97.05%	25.5

4 Introduction

The algorithm was written in VC++ 6.0, the time cost of each plate character extraction varies in range 10ms~20ms. Through experiments on more than 30,000 images, the accurate rate reaches more than 99.2%. The algorithm is fast and accurate enough for practical use. The methods bring a significant contribution to the higher accuracy and improved robustness of LPR system that is developed by our lab and now in product use.

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